IT326

DATA MAINING PROJECT

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Dataset

We applied data mining techniques on Bank Marketing Dataset, which consists of:

- Rows: 41,188
- 21 attributes
- Class label “y” (yes/no) - whether the client subscribed to the term deposit

The Problem

Predicting whether a client will subscribe to a term deposit. Using customer information and data from past marketing campaigns, we seek to understand which factors influence this decision.

Data Pre-processing

Data Cleaning

• No explicit missing values were found in the dataset.
• Ambiguous values (such as "unknown") in categorical columns were addressed.
• These ambiguous values were replaced with the Mode (most frequent value) to ensure data consistency.

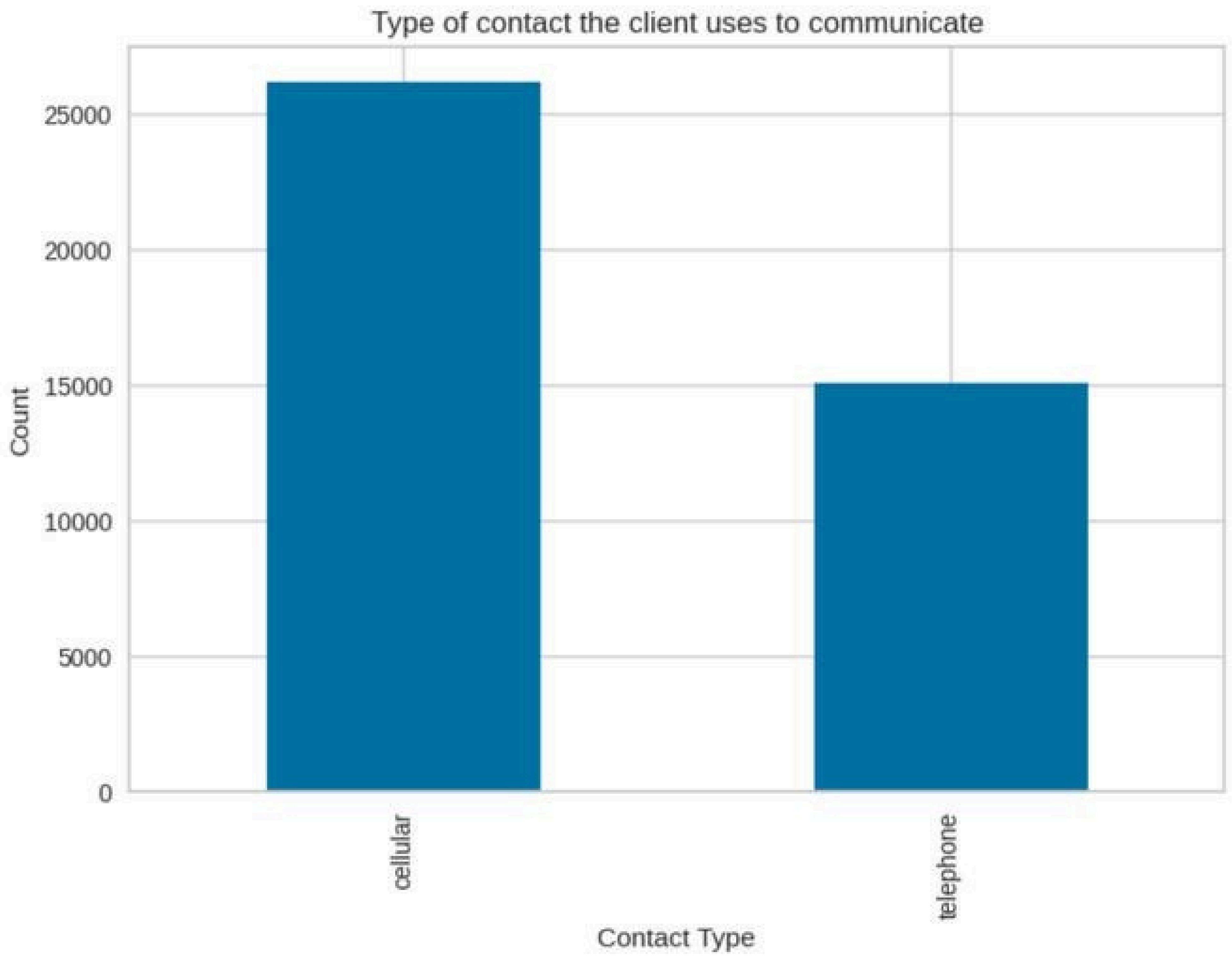
Data Pre-processing

Data Transformation

- The One-Hot Encoding technique was applied to text features.
- The goal: Convert all text columns into a binary numerical format (0s and 1s).
- This transformation is essential for making the data processable by machine learning models.

Analysis

Shows the distribution of contact methods (Cellular/ Telephone).



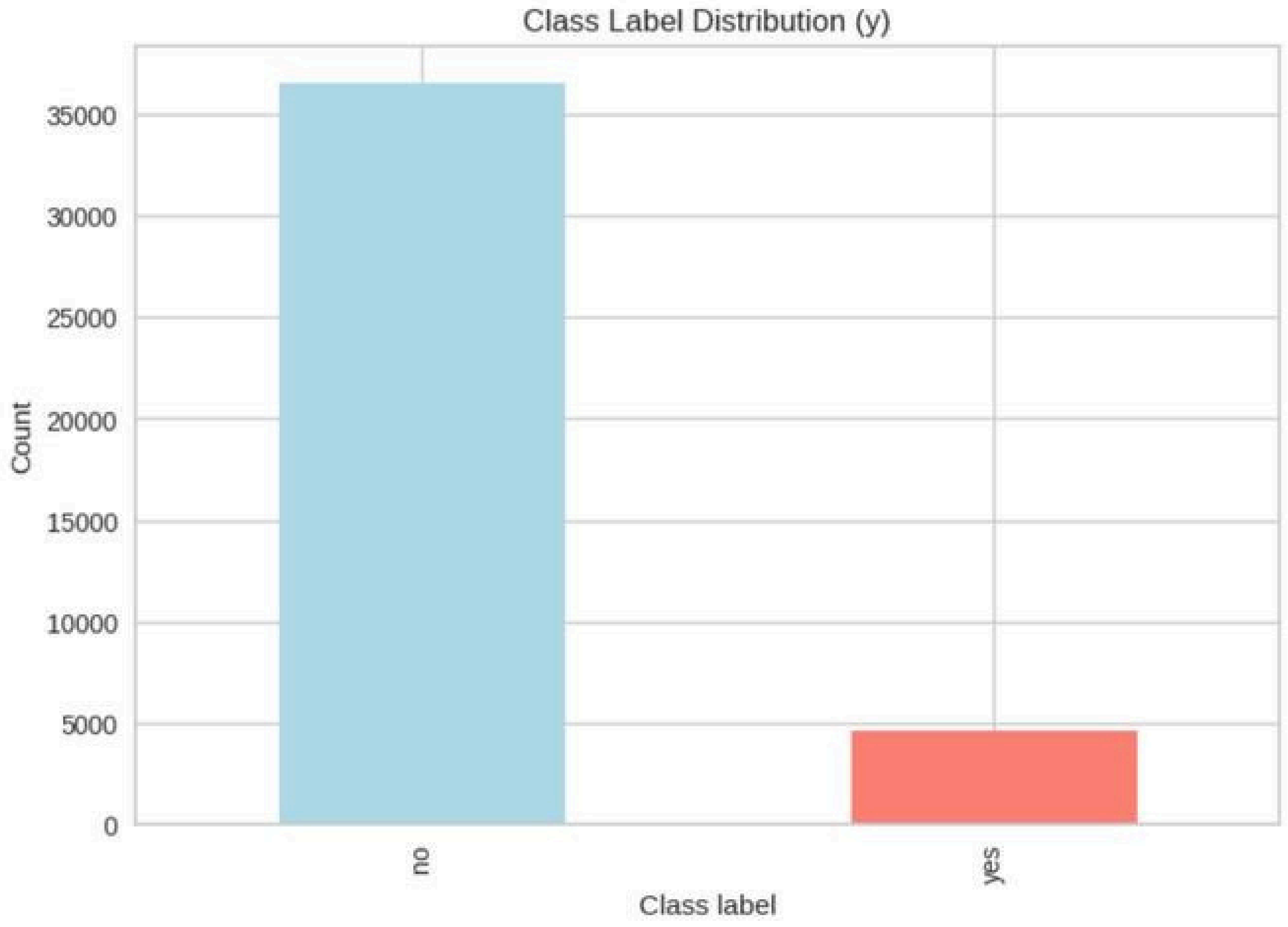
Data Pre-processing

Class label Preparation & Splitting

- The class label (y) was converted to binary numerical values (1 and 0).
- The final dataset was split into Training and Testing sets.
- The splitting goal: Fairly evaluate model performance on new, unseen data.

Analysis

Imbalanced distribution of the target variable. The 'No' category far exceeds 'Yes'.



Classification

What we did:

- Applied a Decision Tree Classifier on the preprocessed dataset.
- Encoded categorical data using Label Encoding.
- Tested three data splits: 80/20, 70/30, 60/40.
- Compared two criteria: Gini and Entropy.

Classification

Main results:

- Entropy performed better than Gini in all partitions.
- Highest accuracy appeared in the 60/40 split.
- The model predicts "no" very well (majority class).
- Weaker performance on "yes" due to class imbalance.

Key insights

The classifier is stable across all train/test ratios.

Most misclassifications are false negatives (predicting "no" instead of "yes").

Entropy is the best overall attribute selection measure.

Findings

Decision Tree works well for this dataset.

Entropy is the best-performing algorithm across all experiments.

Clustering

What we did:

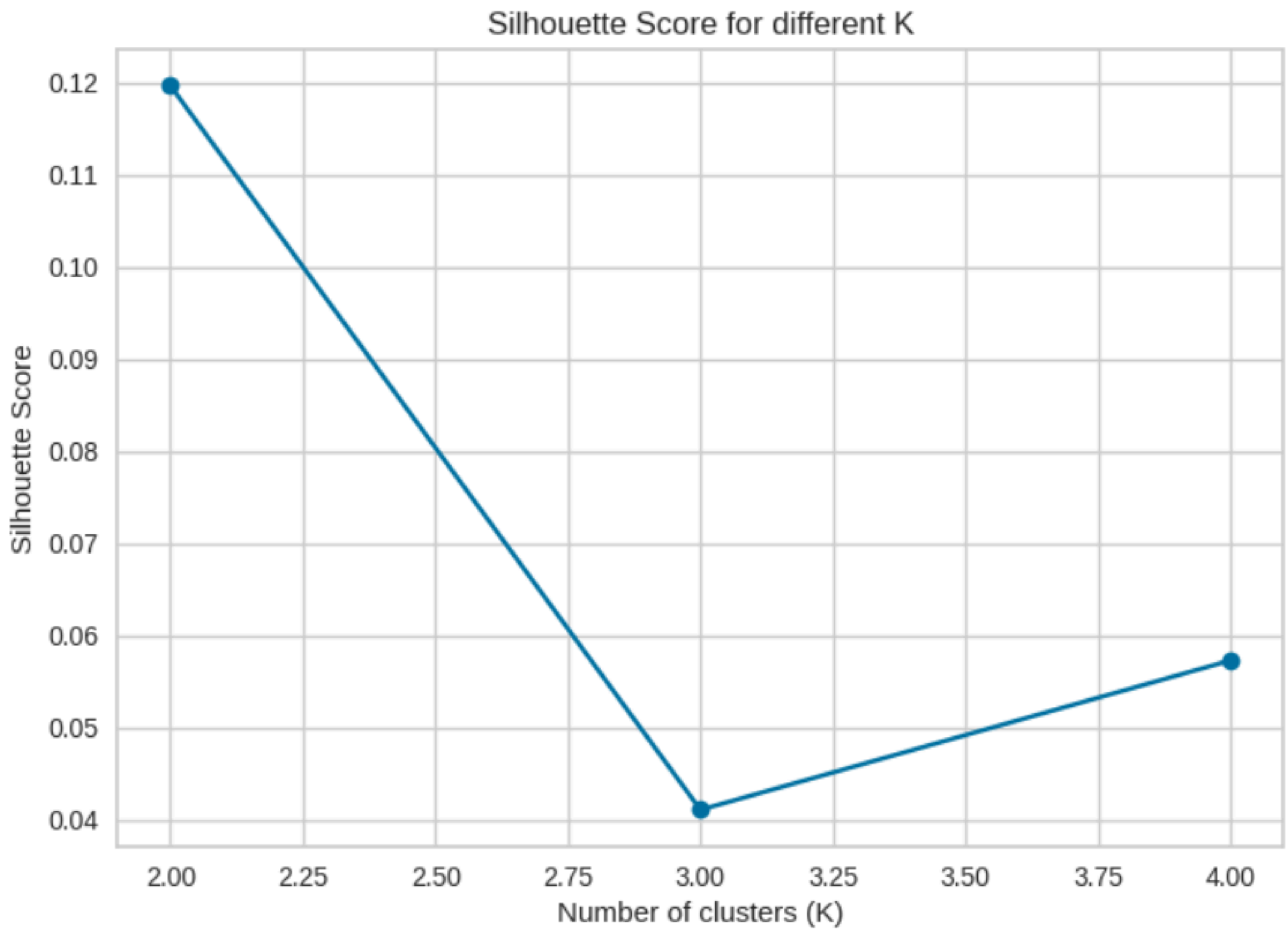
- Applied K-Means with K = 2, 3, 4.
- Calculated Silhouette Score and Within-Cluster Sum of Squares (WCSS).
- Visualized clusters with scatter plots.
- Compared how cluster quality changes when K increases.

Clustering

Main results:

- K = 2 achieved the highest Silhouette Score with the clearest separation.
- WCSS decreases as K increases, but cluster quality does not improve.
- K = 3 and K = 4 show more overlap and weaker group structure.
- Scatter plots confirm that K = 2 forms the most distinct clusters.

Clustering



Silhouette Score for different K
Silhouette Score
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
2.00
2.25
2.50
2.75
3.00
3.25
3.50
3.75
4.00
Number of clusters (K)

Key insights

The dataset is imbalanced, so higher K values mix the groups.

K = 2 provides the most distinct clusters overall.

Silhouette Score confirms that K = 2 is the optimal choice.

Findings

K-Means works reasonably well for this dataset.

K = 2 is the optimal number of clusters, showing the best separation and highest Silhouette Score.

THANK YOU

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